

Neural Network Aided Reconstruction of Ionospheric Tomography for Limited GNSS Measurements

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Introduction

The ionospheric electron densities can be monitored by observations of Total Electron Contents (TEC) from Global Navigation Satellite System (GNSS) measurements, especially in the form of three-dimensional profiles using Computerized Ionospheric Tomography (CIT). Previous works have studied on reconstruction of electron density profiles from slanted TEC measurements with inverse calculation [1].

However, the concept of CIT includes uncertainties originated from the measurements and the inversion calculation. Due to the limitations of observation quantity and geometric structures of GNSS measurements, the inversion process in CIT yields ill-posed problem. Furthermore, the limitations of measurements make it hard to represent temporal dynamics in ionosphere [2][3].

In this work, to overcome these challenges, we will train the neural network to reflect dynamical and statistical model of ionospheric electron density profiles from large amount of measurements. By applying trained neural network, highly accurate tomographic results will be obtained from those under relatively sparse measurements.

Data

GNSS signal measurements ($L_1 = 1575.42$ MHz and $L_2 = 1227.60$ MHz) were obtained from GPS Earth Observation Network (GEONET). Obtained GNSS measurements were input into Long-Term Ionospheric Anomaly Monitor (LTIAM) to estimate Slant TEC (STEC) which is an integrated value of the ionospheric electron density along the raypath between the satellite and the station [4]. To validate the results, we used NmF2 data, which is maximum electron density measurements from ionosonde in Japan.

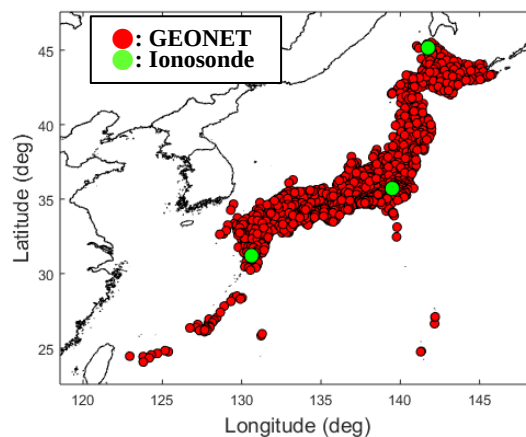


Figure 1 GEONET and ionosonde sites in Japan

Methodology

A. Computerized Ionospheric Tomography (CIT)

The ionospheric delay in GNSS measurement, which can be converted to STEC, can be defined as the integrated value

of the ionospheric electron density, as given by,

$$STEC \approx \sum_{m=1}^M \alpha_m N(\vec{r}, t) \quad (1)$$

where m denote a sampling point in altitude, α is a geometric matrix that contains geometric information on the rays for the defined voxels and $N(\vec{r}, t)$ indicate the electron density at the observation time t and ray vector $\vec{r}$.

To obtain electron density values from the equation (1), iterative numerical methods called Multiplicative Algebraic Reconstruction Technique (MART) is commonly used as follow, equation (2),

$$N_j^{k+1} = N_j^k \left(\frac{STEC_i}{\sum_m \alpha_{im} N_j^k} \right)^{\frac{\lambda \alpha_{ij}}{\sum \alpha_{ij}^2}} \quad (2)$$

where N_j^k is the k -th iterated electron density solution and λ is a relaxation parameter [1].

B. Ionospheric Tomography Neural Network Model

In this study, we considered a three dimensional space over the latitude of 25 – 50° N, the longitude of 125 – 150° E, and the altitude of 90 – 1,000 km. Within this space, as described in Fig 2, electron density profiles from 80 randomly selected stations were obtained as input dataset and those from all 1298 stations were obtained as ground truths for neural network. By testing the neural network for electron density profiles from a different input dataset, we verified the performance of trained neural network and also validated the tested results with NmF2 values measured by ionosonde sites.

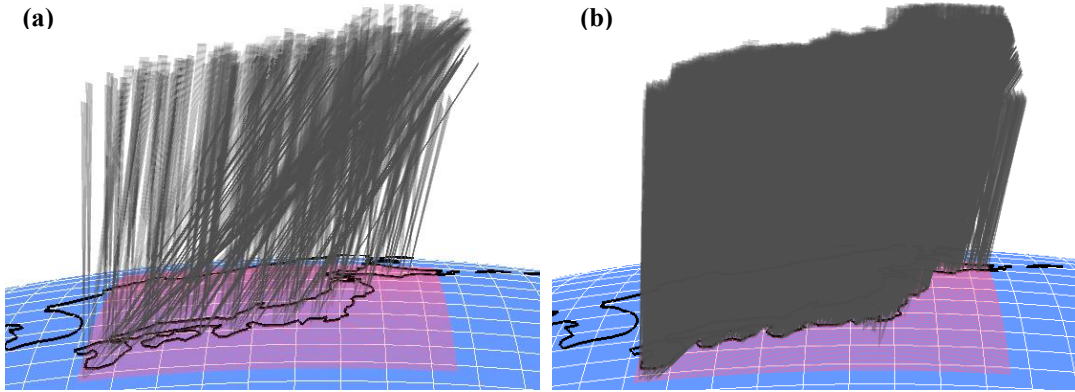


Figure 2 GNSS signal rays for (a) 80 randomly selected stations and (b) all 1298 stations in GEONET

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